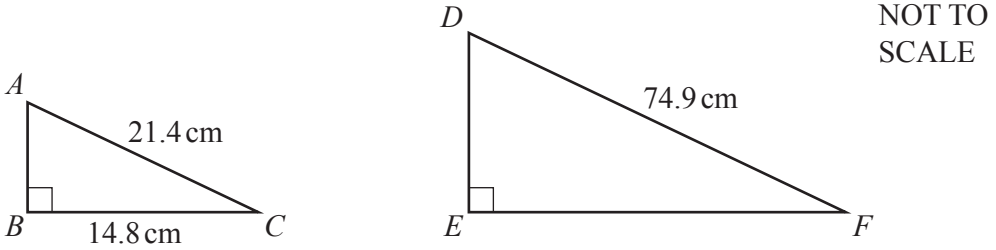
0580_w21_qp_32_Q8

8 **(a)**



Right-angled triangles *ABC* and *DEF* are similar.

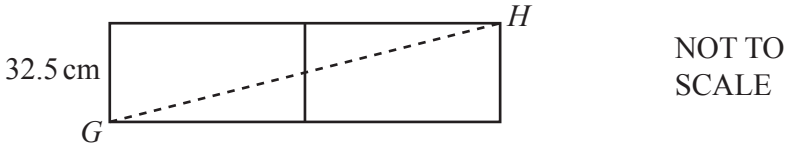
(i) Calculate *EF*.

EF = ... cm [2]

(ii) Calculate angle *BCA*.

Angle *BCA* = ... [2]

(b) The diagram shows two congruent rectangular tiles placed together.



The width of each tile is 32.5 cm and *GH* = 84.5 cm.

Find the length of each tile.

..... cm [4]

0580_w21_qp_32_Q8

(c) Town B is 72 km from town A on a bearing of 058°.
Town C is 60 km due east of town B.

(i) Using a scale of 1 cm to represent 12 km, complete the scale drawing to show the positions of town B and town C.

Scale: 1 cm to 12 km

[3]

(ii) Measure the bearing of town C from town A.

..... [1]